

T1

$$x^2 = y^2$$

a) $x^2 = y^2 \Leftrightarrow y = \pm x, x \in \mathbb{R} \Rightarrow$ в каждой точке (кроме $x=0$) выбираем из x и $-x \rightarrow$ бесконечного много функций

б) $y = x, y = -x, y = |x|, y = -|x| \Rightarrow 4$ функции

в) $y = x, y = |x| \rightarrow 2$ ф-и

г) $y = x, \rightarrow 1$ ф-и

$$\sqrt{3.64} \quad (2) \quad x \cdot \cos y + y \cos u + u \cos x = 1, (0, 1, 0)$$

$$x=0, y=1: 0 \cdot \cos 1 + 1 \cdot \cos u + u \cos 0 = 1$$

$$u + \cos u = 1 \rightarrow F(0, 1, 0) = 0, F'_u(0, 1, 0) = 1 \rightarrow$$

$$u'_x = \cos y - y u'_x \sin u + u'_x \cos x - u \sin x = 0; \text{ вын. усл. Th. 0 не вын. ф.}$$

$$x=0, y=1, u=0 \rightarrow \cos 1 + 0 + u'_x - 0 = 0 \Rightarrow u'_x = -\cos 1$$

$$u'_y: -x \sin y + \cos u - y u'_y \sin u + u'_y \cos x = 0, x=0, y=1, u=0 \rightarrow 0 + 1 - 0 + u'_y = 0 \Rightarrow u'_y = -1$$

$$\sqrt{3.64} \quad (2) \quad x^3 + 2y^3 + u^3 - 3xyu + 2y - 3 = 0, \#$$

$$a) (1, 1, 1), \delta) (1, 1, -2)$$

$$x=1, y=1: 1 + 2 + u^3 - 3u + 2 - 3 = 0 \rightarrow u^3 - 3u + 2 = 0$$

$$F'_u = 3u^2 - 3xy$$

$$a) F(1, 1, 1) = 1 - 3 + 2 = 0 \rightarrow F(1, 1, 1) = 0; F'_u(1, 1, 1) = 0 \rightarrow$$

не вын. усл. Th. 0 не вын. ф-и

$$8) F(1, 1, -2) = 1 + 2 - 8 + 6 + 2 - 3 = 0$$

$$F'_u(1, 1, -2) = 12 - 3 \neq 0$$

блнн. ycn. Th. o неевн. ф-ии

$$u'_x: 3x^2 + 0 + 3u'_x x^2 - 3y u - 3xy u'_x + 0 - 0 = 0$$

$$x=1, y=1, u=-2: 3 + 12u'_x + 6 - 3u'_x = 0 \rightarrow$$

$$u'_y: 0 + 6y^2 + 3u'_y u^2 - 3x u - 3xy u'_y + 2 - 0 = 0$$

$$x=1, y=1, u=-2: u'_y = -\frac{14}{9}$$

$$[du = -dx - \frac{14}{9}dx]$$

$$\sqrt{3.63} \quad f(x-y; y-z; z-x) = 0$$

$$f'_u(dx-dy) + f'_v(dy-dz) + f'_w(dz-dx) = 0$$

$$f'_u dx - f'_u dy + f'_v dy - f'_v dz + f'_w dz - f'_w dx = 0$$

$$[dz = \frac{(f'_u - f'_w)dx + (f'_v - f'_u)dy}{f'_v - f'_w}]$$

$$\sqrt{3.75} \quad \begin{cases} e^{u+v} + x(u'_x + v'_x)e^{u+v} + 2u'_x v + 2u v'_x = 0 \\ x e^{u+v}(u'_y + v'_y) + 2u'_y v + 2u v'_y = 0 \\ y e^{u+v}(u'_x + v'_x) - (u'_x(1+v) + u v'_x)(1+v) = 2 \\ e^{u+v} + e^{u+v}(u'_y + v'_y) - (u'_y(1+v) + u v'_y)(1+v) = 0 \end{cases}$$

$$x=1, y=2: \begin{cases} 1 + 2u'_x + v'_x = 0 \\ u'_y + v'_y = 0 \\ 2(u'_x - v'_x) - 2u'_x = 2 \\ 1 + 2(u'_x - v'_x) - 2u'_y = 0 \end{cases} \quad \begin{cases} u'_x = 0 \\ u'_y = -\frac{1}{3} \\ v'_x = -1 \\ v'_y = \frac{1}{3} \end{cases}$$

1/4.43

$$5) u^3 + 2yu + xy = 0, u(1; -1) = -1$$

$$3u^2 du + 2dy u + 2y dx + x dy = 0$$

$$x=1, y=-1, u=-1: 3du - 2dy - 2dx - dx + dy = 0 \rightarrow du = dx + dy$$

$$6u du^2 + 3u^2 d^2 u + 2du dy + 2y d^2 u + 2dy du + dx dy + dx dy = 0$$

$$-6dx^2 + -6dy^2 - 12dx dy + 3d^2 u + 4dy^2 + 4dx dy - 2d^2 u + 2dx dy = 0$$

$$d^2 u = 6dx^2 + 2dy^2 + 6dx dy$$

1/4.46

$$f(x+u, y+u) = 0$$

$$f'_1 = f'_{x+u}, f'_2 = f'_{y+u}, f''_{11}, f''_{12}, f''_{22} - \text{аналогично}$$

$$d'_1 dx + f'_1 du + f'_2 dy + f'_2 du = 0 \rightarrow du = -[f'_1 dx + f'_2 dy] \cdot [f'_1 + f'_2]^{-1}$$

$$dx + du = \frac{f'_1 dx + f'_2 dx - f'_1 dx - f'_2 dy}{f'_1 + f'_2} = \frac{f'_2(dx - dy)}{f'_1 + f'_2}$$

$$dy + du = \frac{f'_1 dy + f'_2 dy - f'_1 dx - f'_2 dy}{f'_1 + f'_2} = -\frac{f'_1(dx - dy)}{f'_1 + f'_2}$$

$$d^2 f = f''_{11} (d(x+u))^2 + 2f''_{12} d(x+u) \cdot d(y+u) + f''_{22} (d(y+u))^2 + f'_1 d^2(x+u) + f'_2 d^2(y+u) = 0$$

$$d^2 u (f'_1 + f'_2) = -[f''_{11} (d(x+u))^2 + 2f''_{12} d(x+u) d(y+u) + f''_{22} (d(y+u))^2]$$

$$d^2 u = -\frac{(dx - dy)^2}{(f'_1 + f'_2)^3} \cdot [f''_{11} \cdot (f'_1)^2 - 2f''_{12} \cdot f'_1 \cdot f'_2 + f''_{22} \cdot (f'_2)^2]$$

√3.105

$$\frac{\partial(x, y, z)}{\partial(r, \varphi, \psi)} = \begin{vmatrix} \cos^p \varphi \cos^q \psi & -r \cos^p \varphi \sin^q \psi & -r q \cos^p \varphi \cos^{q-1} \psi \sin \psi \\ \sin^p \varphi \cos^q \psi & r \sin^p \varphi \cos^q \psi & -r q \sin^p \varphi \cos^{q-1} \psi \sin \psi \\ \sin^q \psi & 0 & q r \sin^q \psi \cos \psi \end{vmatrix}$$

$$= \sin^{q+1} \psi \cdot r^2 p \cos^q \psi \sin^{p-1} \varphi \cos^{p-1} \varphi (\sin^2 \varphi + \cos^2 \varphi) + \\ + p q r^2 \sin^{q-1} \psi \cos^{2q+1} \psi \cdot \cos^{p-1} \varphi \sin^{p-1} \varphi \cdot (\cos^2 \varphi + \sin^2 \varphi) = \\ = p q r^2 (\sin^p \varphi \cos^p \varphi)^{p-1} \cdot (\cos^2 \psi)^{2q-1} (\sin^2 \psi)^{q-1}$$

T3 $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $u = e^x \cos y$
 $v = e^x \sin y$

1) $J = \begin{vmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{vmatrix} = e^{2x} > 0$

Но $u(x, y) = u(x + \pi; y + 2\pi)$ отображение не биективное,
 $v(x, y) = v(x + \pi; y + 2\pi)$ в силу периодичности

2) $u = \operatorname{Re}(e^{x+iy})$, $v = \operatorname{Im}(e^{x+iy})$

e^z принимает все значения кроме 0 $\rightarrow$

$$f(\mathbb{R}^2) = \mathbb{R}^2 \setminus \{0, 0\}$$

T4 $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $x = r \cos \varphi$
 $y = r \sin \varphi$

$$\begin{cases} 1 = r'_x \cos \varphi + r(-\sin \varphi) \varphi'_x \\ 0 = r'_y \cos \varphi + r(-\sin \varphi) \varphi'_y \\ 0 = r'_x \sin \varphi + r \cos \varphi \varphi'_x \\ 1 = r'_y \sin \varphi + r \cos \varphi \varphi'_y \end{cases}$$

$$r'_x = \frac{\Delta r_1}{\Delta_1} = \cos \varphi$$

$$\varphi'_x = \frac{\Delta \varphi_1}{\Delta_1} = -\frac{\sin \varphi}{r}$$

$$r'_y = \frac{\Delta r_2}{\Delta_2} = \sin \varphi$$

$$\varphi'_y = \frac{\Delta \varphi_2}{\Delta_2} = \frac{\cos \varphi}{r}$$

$$\begin{cases} r'_x \cos \varphi + r(-\sin \varphi) \varphi'_x = 1 \\ r'_x \sin \varphi + r \cos \varphi \varphi'_x = 0 \end{cases}$$

$$\begin{cases} r'_y \cos \varphi - \varphi'_y r \sin \varphi = 0 \\ r'_y \sin \varphi + \varphi'_y r \cos \varphi = 1 \end{cases}$$

√3.86 $x \frac{\partial u}{\partial y} - y \frac{\partial u}{\partial x} = 0$

$$x u'_y - y u'_x = 0, \quad x = r \cos \varphi, \quad y = r \sin \varphi$$

$$u'_x = u'_r \cdot r'_x + u'_\varphi \cdot \varphi'_x = u'_r \cos \varphi - u'_\varphi \frac{\sin \varphi}{r} \geq T4$$

$$u'_y = u'_r \cdot r'_y + u'_\varphi \cdot \varphi'_y = u'_r \sin \varphi + u'_\varphi \frac{\cos \varphi}{r}$$

$$r \cos \varphi (u'_r \sin \varphi + u'_\varphi \frac{\cos \varphi}{r}) - r \sin \varphi (u'_r \cos \varphi - u'_\varphi \frac{\sin \varphi}{r}) = 0$$

$$\rightarrow u'_\varphi = 0 \rightarrow \underline{u(r, \varphi) = f(r) = f(x^2 + y^2)}$$

√88 $\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = 0$, $u = x + y$
 $v = x - y$

$$u'_x = 1, u'_y = 1$$

$$v'_x = 1, v'_y = -1$$

$$\begin{cases} z'_x = z'_u u'_x + z'_v v'_x = z'_u + z'_v \\ z'_y = z'_u u'_y + z'_v v'_y = z'_u - z'_v \end{cases} \rightarrow$$

$$z'_u + z'_v - z'_u + z'_v = 0 \rightarrow z'_v = 0$$

$$z = f(u) = f(x+y)$$

$$\sqrt{91} \left| (y-z) \frac{\partial z}{\partial x} + (y+z) \frac{\partial z}{\partial y} = 0 \right.$$

$$u = y - z, v = y + z, x(u, v)$$

$$dx = x'_u du + x'_v dv = x'_u dy - x'_u dz + x'_v dy + x'_v dz$$

$$= (x'_u + x'_v) dy + (x'_v - x'_u) dz \rightarrow$$

$$dz = (x'_v - x'_u)^{-1} dx + (x'_u + x'_v) (x'_u - x'_v)^{-1} dy$$

$$u (x'_v - x'_u)^{-1} + v (x'_u + x'_v) (x'_u - x'_v)^{-1} = 0 \rightarrow$$

$$\rightarrow u = v (x'_u + x'_v) \rightarrow \left(\frac{u}{v} = x'_u + x'_v \right)$$

$$\sqrt{51(2)} \left| x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 0 \right.$$

$$x = r \cos p$$

$$y = r \sin p$$

$$u'_x = u'_r \cdot \cos p - u'_p \cdot \frac{\sin p}{r}$$

$$r'_x = \cos p$$

$$u'_y = u'_r \sin p + u'_p \cdot \frac{\cos p}{r}$$

$$r'_y = \sin p$$

$$p'_x = -\frac{\sin p}{r}, p'_y = \frac{\cos p}{r}$$

$$u''_{xx} = (u'_x)'_r \cdot r'_x + (u'_x)'_p \cdot p'_x =$$

$$= u''_{rr} \cos^2 p - 2 \frac{\sin p \cos p}{r} u''_{rp} + \frac{\sin^2 p}{r^2} u''_{pp} + \frac{\sin^2 p}{r} u'_r + 2 \frac{\cos p \sin p}{r^2} u'_p$$

$$u''_{yy} = u''_{rr} \sin^2 p + \frac{2 \cos p \sin p}{r} u''_{rp} + u''_{pp} \frac{\cos^2 p}{r^2} + u'_r \frac{\cos^2 p}{r} - 2 \frac{\sin p \cos p}{r} u'_p$$

$$u''_{yx} = u''_{rr} \sin p \cos p + u''_{rp} \left(\frac{\cos^2 p - \sin^2 p}{r} \right) - u''_{pp} \frac{\sin p \cos p}{r^2} + u'_p \left(\frac{\sin^2 p - \cos^2 p}{r^2} \right) - u'_r \frac{\sin p \cos p}{r}$$

$$u''_{xx} r^2 \cos^2 p + u''_{yx} r^2 2 \sin p \cos p + u''_{yy} r^2 \sin^2 p = 0$$

$$u''_{rr} r^2 (\cos^4 p + \sin^4 p + 2 \sin^2 p \cos^2 p) = 0$$

$$u''_{rr} \cdot r^2 = 0$$

$$\boxed{r^2 \cdot \frac{\partial^2 u}{\partial r^2} = 0}$$

$$\sqrt{52(1)} \left| \frac{\partial^2 z}{\partial t^2} = a^2 \frac{\partial^2 z}{\partial x^2}, u = x - at, v = x + at \right.$$

$$z''_{tt} = a^2 z''_{xx}, u = x - at, v = x + at$$

$$z'_x = z'_u u'_x + z'_v v'_x = z'_u + z'_v$$

$$z'_t = z'_u u'_t + z'_v v'_t = a(-z'_u + z'_v)$$

$$z''_{xx} = z''_{uu} + z''_{vv} + z''_{uv} + z''_{vu} = z''_{uu} + 2z''_{uv} + z''_{vv}$$

$$z''_{tt} = a [(-z''_{uu} + z''_{vv}) (a - a) + (2 - z''_{uv} + z''_{vv}) \cdot a] =$$

$$= a^2 (z''_{uu} - 2z''_{uv} + z''_{vv})$$

$$a^2 (z''_{uu} - 2z''_{uv} + z''_{vv}) = a^2 (z''_{uu} + 2z''_{uv} + z''_{vv}) \rightarrow$$

$$\rightarrow \boxed{z''_{uv} = 0}$$

Т5

а) гл, 5.9. - пример

б) нет

в) гл, пример гл 5.9

№2.2

$$u = 3x^2y + y^3 - 12x - 15y + 3$$

$$u'_x = 6xy - 12 = 0 \rightarrow xy = 2 \rightarrow (2, 1), (-2, -1)$$

$$u'_y = 3x^2 + 3y^2 - 15 = 0 \rightarrow x^2 + y^2 = 5 \rightarrow (1, 2), (-1, -2)$$

$$u''_{xx} = 6y$$

$$u''_{yy} = 6y$$

$$u''_{xy} = 6x$$

$$d^2f = 6[y(dx^2 + dy^2) + 2xdxdy]$$

$$1) (1, 2): d^2f = 12(dx^2 + dy^2 + dxdy), \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}$$

$$\Delta_1 > 0 \\ \Delta_2 > 0 \rightarrow \text{лок. мин.}$$

$$2) (-1, -2): d^2f = -12(dx^2 + dy^2 + dxdy), \begin{bmatrix} -1 & -1/2 \\ -1/2 & -1 \end{bmatrix}$$

$$\Delta_1 < 0 \leftrightarrow \text{лок. макс} \\ \Delta_2 > 0$$

$$3) (2, 1): d^2f = 6(dx^2 + dy^2 + 4dxdy), \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\Delta_1 > 0 \rightarrow \text{нет лок. экстр.} \\ \Delta_2 < 0$$

$$4) (-2, -1): d^2f = -6(dx^2 + dy^2 + 4dxdy), \begin{bmatrix} -1 & -2 \\ -2 & -1 \end{bmatrix}$$

$$\Delta_1 < 0 \rightarrow \text{нет лок. экстр} \\ \Delta_2 < 0$$

$$u(1, 2) = -25 - \text{мин.}$$

$$u(-1, -2) = 31 - \text{макс}$$

$$\text{№3} \quad u = x^4 + y^4 - 2x^2$$

$$u'_x = 4x^3 - 4x = 0 \quad x(x^2 - 1) = 0 \quad (0, 0), (1, 0), (-1, 0)$$

$$u'_y = 4y^3 = 0 \quad y = 0$$

$$u''_{xx} = 12x^2 - 4 \quad d^2u = (12x^2 - 4)dx^2 + 12y^2dy^2$$

$$u''_{yy} = 12y^2 \quad (0, 0): d^2u = -4dx^2 - \text{отриц. полугонр}$$

$$u''_{xy} = 0 \quad \Delta u(0, 0) = u(\Delta x, \Delta y) - u(0, 0) = (\Delta x)^4 + (\Delta y)^4 - 2(\Delta x)^2$$

$$= (\Delta x)^2((\Delta x)^2 - 2) + (\Delta y)^4$$

$$\text{Если } 0 < |\Delta x| < \sqrt{2}, \Delta y = 0 \rightarrow \ominus, \text{Если } \Delta x = 0, \Delta y \neq 0 \rightarrow \oplus$$

Нет лок. экстр

$$(\pm 1, 0): d^2u = 8dx^2 - \text{полож. полугонр.}$$

$$\Delta u(\pm 1, 0) = u(\pm 1 + \Delta x, \Delta y) - u(\pm 1, 0) =$$

$$= (\pm 1 + \Delta x)^4 + (\Delta y)^4 - 2(\pm 1 + \Delta x)^2 + 1 =$$

$$= (\Delta x)^2(2 \pm \Delta x)^2 + (\Delta y)^4 \geq 0$$

$$\text{Если } \Delta u = 0 \rightarrow \Delta y = 0, \Delta x = 0 \rightarrow \text{Зстрозим лок. мин}$$

$$u(\pm 1, 0) = -1$$

$$\begin{aligned} \text{N13.1} \quad u'_x &= 2x - y + 1 = 0 \\ u'_y &= 2y - x = 0 \quad \rightarrow \left(-\frac{2}{3}, -\frac{1}{3}, -1\right) \\ u'_z &= 2z + 2 = 0 \end{aligned}$$

$$\begin{aligned} u''_{xx} &= 2 & u''_{xz} &= 0 \\ u''_{xy} &= 2 & u''_{xy} &= -1 \rightarrow \\ u''_{zz} &= 2 & u''_{yz} &= 0 \end{aligned}$$

$$d^2u = 2dx^2 + 2dy^2 + 2dz^2 - 2dx dy$$

$$d^2u \left(-\frac{2}{3}, -\frac{1}{3}, -1\right) = 2(dx^2 + dy^2 - dz^2 - dx dy)$$

$$\begin{bmatrix} 1 & 1/2 & 0 \\ 1/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} \Delta_1 > 0 \\ \Delta_2 > 0 \\ \Delta_3 > 0 \end{matrix} \rightarrow \boxed{\text{локл. экстр.}} \\ u\left(-\frac{2}{3}, -\frac{1}{3}, -1\right) = -\frac{1}{3}$$

$$\begin{aligned} \text{N18.1} \quad 2x dx + 2y dy + 2u du + 2dx - 2dy + 4du &= 0 \\ (x+1)dx + (y-1)dy &= -(u+2)du \quad (*) \\ d\left[(x+1)dx + (y-1)dy\right] (-u-2)^{-1} &, x=-1, y=1- \\ 1+1+u^2-2-2+4u-3=0 &\rightarrow u^2+4u-5=0 \quad \text{с.т.т.} \\ \rightarrow u_1=1, u_2=-5 \end{aligned}$$

Дифференцируя (*): $dx^2 + dy^2 = -(u+2)d^2u - du^2$

$$d^2u = -(dx^2 + dy^2)(u+2)^{-1}$$

$$(-1, 1, 1): d^2u = -\frac{1}{3}(dx^2 + dy^2) \quad \text{отриц. опре} \rightarrow \text{локл. макс.}$$

$$(-1, 1, 5): d^2u = \frac{1}{3}(dx^2 + dy^2) \quad \text{отриц. опре} \rightarrow \text{локл. экстр.}$$

$$\begin{aligned} \text{N19} \quad u &= xy, x+y-1=0 \\ y &= 1-x, u = x(1-x) = x-x^2 \rightarrow \max \text{ в т. } \left(\frac{1}{2}, \frac{1}{2}\right) \\ \text{ген. макс } u\left(\frac{1}{2}, \frac{1}{2}\right) &= \frac{1}{4} \end{aligned}$$

$$\begin{aligned} \text{N21} \quad u &= 1-4x-8y, x^2-8y^2=8 \\ L &= 1-4x-8y + \lambda(x^2-8y^2-8) \end{aligned}$$

$$\begin{cases} L'_x = -4 + 2\lambda x = 0 \\ L'_y = -8 - 16\lambda y = 0 \\ x^2 - 8y^2 = 8 \end{cases}$$

$$\lambda = 0 \text{ не подходит} \rightarrow x = \frac{2}{\lambda}, y = -\frac{1}{2\lambda}$$

$$\frac{4}{\lambda^2} - \frac{2}{\lambda^2} = 8 \rightarrow \lambda = \pm \frac{1}{2} \rightarrow x = \pm 4, y = \mp 1$$

$$\begin{aligned} L''_{xx} &= 2\lambda \\ L''_{yy} &= -16\lambda \\ L''_{xy} &= 0 \end{aligned}$$

$$d^2L = 2\lambda(dx^2 - 8dy^2)$$

генер. экстр. сб: $2x dx - 16y dy = 0 \rightarrow dx = 8 \frac{y}{x} dy$

$$\lambda = \frac{1}{2}, x=4, y=-1: d^2L = 4dy^2 - 8dy^2 = -4dy^2 \text{ — отриц. опре.}$$

$$\rightarrow u(4, -1) = -7 \text{ — макс}$$

$$\lambda = -\frac{1}{2}, x=-4, y=1: d^2L = -4dy^2 + 8dy^2 = 4dy^2 \text{ — неопред. опре.}$$

$$\underline{u(-4, 1) = 9 \text{ — мин}}$$

Т6

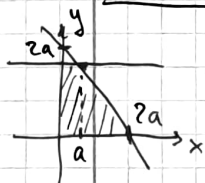
а) Нет, пример в 23.

б) Нет: $f(0, y) = \begin{cases} 1, y \in \mathbb{Q} \\ 0, y \in \mathbb{R}/\mathbb{Q} \end{cases}$ - не упр.,
при $x=0$

√79

$G: x=0, y=0, y=a, x+y=2a$

$$\int_0^a dx \int_0^a f(x, y) dy + \int_a^{2a} dx \int_0^{2a-x} f(x, y) dy = \int_0^a dy \int_0^{2a-y} f(x, y) dx$$



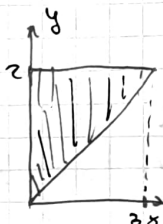
√83

$$\int_{-1}^1 dx \int_{x^2}^{2x^2-1} f(x, y) dy$$

$$\begin{aligned} \int_{-1}^1 dx \int_{x^2}^{2x^2-1} f(x, y) dy &= \int_0^1 dy \int_{-\sqrt{y}}^{-\sqrt{y+1}} f(x, y) dx + \int_0^1 dy \int_{\sqrt{y+1}}^{\sqrt{y}} f(x, y) dx + \\ &+ \int_{-1}^0 dy \int_{-\sqrt{y+1}}^{\sqrt{y+1}} f(x, y) dx \end{aligned}$$

√85 $\int_0^a dx \int_x^a (a^2 - y^2)^\alpha dy, a > 0, \alpha > 0$

$$\begin{aligned} \int_0^a dx \int_x^a (a^2 - y^2)^\alpha dy &= \int_0^a dy \int_0^y (a^2 - y^2)^\alpha dx = \\ &= \int_0^a y (a^2 - y^2)^\alpha dy = -\frac{1}{2} \int_0^a (a^2 - y^2)^\alpha d(a^2 - y^2) = \\ &= -\frac{1}{2} \cdot \frac{(a^2 - y^2)^{\alpha+1}}{\alpha+1} \Big|_0^a = \frac{a^{2(\alpha+1)}}{2(\alpha+1)} \end{aligned}$$



√50

1) $\iint_G (x \sin y + y \cos x) dx dy, G = [0; \frac{\pi}{2}] \times [0; \frac{\pi}{2}]$

$$\begin{aligned} I &= \int_0^{\pi/2} dx \int_0^{\pi/2} (x \sin y + y \cos x) dy, I_0 = (-x \cos y + \frac{1}{2} y^2 \cos x) \Big|_0^{\pi/2} \\ &= 0 + \frac{\pi^2}{8} \cos x + x \\ I &= \int_0^{\pi/2} \left(\frac{\pi^2}{8} \cos x + x \right) dx = \left(\frac{\pi^2}{8} \sin x + \frac{1}{2} x^2 \right) \Big|_0^{\pi/2} = \frac{\pi^2}{4} \end{aligned}$$

8) $\iint_G \sqrt{x-y} dx dy, G = \{ \frac{y}{5} \leq y \leq x, 1 \leq y \leq 4 \}$

$$I = \iint_G \sqrt{x-y} dx dy = \int_1^4 dy \int_y^{\frac{5}{4}y} \sqrt{x-y} dx$$

$$I_0 = \frac{2}{3} (x-y)^{3/2} \Big|_y^{\frac{5}{4}y} = \frac{2}{3} \left(\frac{1}{4} y \right)^{3/2}$$

$$I = \frac{2}{3} \cdot \frac{1}{8} \cdot \int_1^4 y^{3/2} dy = \frac{2}{3} \cdot \frac{1}{8} \cdot \frac{2}{5} (y^{5/2} - 1) = \frac{1}{30} (32 - 1) = \frac{31}{30}$$

√133

$$\int_0^4 dz \int_{\frac{3-3z}{4}}^{\frac{2-2z}{3}-\frac{z}{2}} \int_0^{\frac{4-2x-\frac{y}{2}}{2}} f(x, y, z) dx$$

$$= \int_0^2 dx \int_0^{3-\frac{3}{2}x} dy \int_0^{4-2x-\frac{y}{2}} f(x, y, z) dz$$

√139

$f(x, y, z) = (1+x+y+z)^{-3}, G$ огранич.

$$\begin{aligned} I &= \iiint_G f(x, y, z) dx dy dz = \iint_S dx dy \int_0^{1-x-y} f(x, y, z) dz = \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} (1+x+y+z)^{-3} dz \\ &= \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} (1+x+y+z)^{-3} dz \end{aligned}$$

$$I_0 = -\frac{1}{2} (1+x+y+z)^{-2} \Big|_0^{1-x-y} = -\frac{1}{2} [2^{-2} - (1+x+y)^{-2}]$$

$$I = \frac{1}{2} \int_0^1 dx \int_0^{1-x} [(1+x+y)^{-2} - 2^{-2}] dy$$

$$I_1 = -\frac{1}{2} (1+x+y)^{-1} \Big|_0^{1-x} - \frac{1}{4} y \Big|_0^{1-x} = -2^{-1} + (1+x)^{-1} - \frac{1}{4}(1-x)$$

$$I = \frac{1}{2} \int_0^1 [-2^{-1} + (1+x)^{-1} - \frac{1}{4}(1-x)] dx = \frac{1}{2} \left(-\frac{1}{2} + \ln 2 - \frac{1}{8} \right) = \frac{1}{2} \ln 2 - \frac{5}{8} = \frac{8 \ln 2 - 5}{16}$$

N175 $Q_n = [0, a]^n \in \mathbb{R}^n, n \geq 2$

$$\int_{Q_n} \sum_{k=1}^n x_k dx = \int_0^a dx_1 \int_0^a dx_2 \int_0^a \dots \int_0^a dx_{n-1} \int_0^a \sum_{k=1}^n x_k dx_n =$$

$$= \int_0^a x_1 dx_1 \int_0^a \dots \int_0^a dx_n + \int_0^a dx_1 \int_0^a x_2 dx_2 \int_0^a \dots \int_0^a dx_n + \dots +$$

$$\int_0^a dx_1 \int_0^a \dots \int_0^a x_n dx_n = n \cdot \int_0^a x_1 dx_1 \int_0^a \dots \int_0^a dx_n = n \cdot a^{n-1} \cdot \frac{a^2}{2} =$$

$$= \frac{n a^{n+1}}{2}$$

N176 $\Pi_n = \{0 \leq x_n \leq x_{n-1} \leq \dots \leq x_2 \leq x_1 \leq a\}$:

$$\int x_1 x_2 \dots x_n dx = \int_0^a x_1 dx \int_0^{x_1} x_2 dx_2 \int_0^{x_2} \dots \int_0^{x_{n-2}} x_{n-1} dx_{n-1} \int_0^{x_{n-1}} dx_n$$

$$= \int_0^a \Pi_n x_1 \int_0^{x_1} x_2 dx_2 \int_0^{x_2} \dots \int_0^{x_{n-2}} \frac{x_{n-1}^3}{2} dx_{n-1} = \dots = \frac{a^{2n}}{(2n)!!}$$

$$n=1: \int_0^a x_1 dx_1 = \frac{a^2}{2}, n=2: \int_0^a \frac{x_1^3}{2} = \frac{a^4}{2 \cdot 4}, n=3: \int_0^a \frac{x_1^5}{2 \cdot 4} = \frac{a^6}{2 \cdot 4 \cdot 6}$$

3) $\iint_G |xy| dx dy, G = \{a^2 \leq x^2 + y^2 \leq 4a^2\}$

$$\int_0^{2\pi} d\varphi \int_a^{2a} r^3 |\cos \varphi \sin \varphi| dr = \frac{15a^4}{4} \cdot 4 \cdot \int_0^{\pi/2} \cos \varphi \sin \varphi d\varphi = 15a^4 \int_0^{\pi/2} \sin \varphi d\varphi = \frac{15}{2} a^4$$

N107 $\iint_G y dx dy, G = \{x^2 + y^2 \leq 2x, x > y\}$

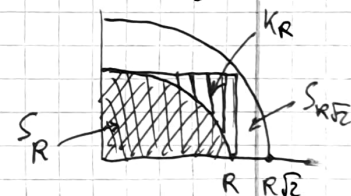
$$x^2 + y^2 \leq 2x \rightarrow (x-1)^2 + y^2 \leq 1, r^2 \leq 2r \cos \varphi$$

$$I = \int_{-\pi/4}^{\pi/4} d\varphi \int_0^{2 \cos \varphi} r^2 \sin \varphi dr + \int_{-\pi/2}^{-\pi/4} d\varphi \int_0^{2 \cos \varphi} r^2 \sin \varphi dr =$$

$$= \frac{8}{3} \left[\int_{-\pi/4}^{\pi/4} \sin \varphi d\varphi + \int_{-\pi/2}^{-\pi/4} \cos^3 \varphi \sin \varphi d\varphi \right] = \frac{8}{3} \cdot \frac{1}{4} \cos^4 \varphi \Big|_{-\pi/2}^{-\pi/4} = \frac{1}{6}$$

N110 $\int_0^{+\infty} e^{-x^2} dx$

$$\iint_{S_R} e^{-(x^2+y^2)} dx dy \leq \iint_{K_R} e^{-(x^2+y^2)} dx dy = \iint_{S_{R\sqrt{2}}} e^{-(x^2+y^2)} dx dy$$



$$\iint_{K_R} = \int_0^R e^{-x^2} dx \cdot \int_0^R e^{-y^2} dy = \left(\int_0^R e^{-x^2} dx \right)^2$$

$$\iint_{S_R} = \int_0^{\pi/2} d\varphi \int_0^R e^{-r^2} r dr = \left\{ \begin{matrix} r^2 = t \\ 2r dr = dt \end{matrix} \right\} = \frac{\pi}{4} \int_0^{R^2} e^{-t} dt = \frac{\pi}{4} (1 - e^{-R^2})$$

Поиск б (1): $\frac{\pi}{4} (1 - e^{-R^2}) \leq \left(\int_0^R e^{-x^2} dx \right)^2 \leq \frac{\pi}{4} (1 - e^{-2R^2})$
 $R \rightarrow \infty \rightarrow \frac{\pi}{4} \leq \left(\int_0^R e^{-x^2} dx \right)^2 \leq \frac{\pi}{4} \rightarrow \int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$

№124

1) $f(x, y) = x + y$, G ограничено линиями $xy = a$, $xy = b$

$y = x$, $y = x - c$, $0 < a < b$, $0 < c$

$xy = u \rightarrow \frac{D(u, v)}{D(x, y)} = \begin{vmatrix} y & x \\ -1 & 1 \end{vmatrix} = y + x = f(x, y)$
 $y - x = v$

$\iint_G du dv = \int_a^b du \int_{-c}^0 dv = (b - a) \cdot c$

5) $f(x, y) = x^4 - y^4$, $G = \{x > 0, 1 \leq xy \leq 2, x^2 - y^2 \leq 2\}$
 $xy = u \rightarrow \frac{D(u, v)}{D(x, y)} = \begin{vmatrix} y & x \\ 2x & -2y \end{vmatrix} = -2(x^2 + y^2)$
 $x^2 - y^2 = v$

$$\frac{x^4 - y^4}{2(y^2 + x^2)} = \frac{x^2 - y^2}{2} = \frac{1}{2} v$$

$$\frac{1}{2} \iint_{G^*} v du dv = \frac{1}{2} \int_1^2 du \int_1^2 v dv = \frac{1}{2} \cdot \frac{3}{2} \int_1^2 du = \frac{3}{4}$$

№144.6 $\iiint_G f(x, y, z) dx dy dz$

$f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$, $G = \{x^2 + y^2 + z^2 \leq z\}$
 $\rightarrow r^2 < r \sin \psi, r < \sin \psi$

$z = r \sin \psi$

$x = r \cos \psi \cos \varphi$, $-\frac{\pi}{2} \leq \psi \leq \frac{\pi}{2}$

$y = r \cos \psi \sin \varphi$, $-\pi \leq \varphi \leq \pi$

$\frac{D(x, y, z)}{D(r, \psi, \varphi)} = r^2 \cos \psi$

$x^2 + y^2 + \left(z - \frac{1}{2}\right)^2 \leq \frac{1}{4}$
 $\int_{-\pi}^{\pi} d\varphi \int_0^{\pi/2} d\psi \int_0^{\sin \psi} r r^2 \cos \psi dr = 2\pi \int_0^{\pi/2} \cos \psi d\psi \int_0^{\sin \psi} r^3 dr =$
 $= 2\pi \int_0^{\pi/2} \cos \psi d\psi \cdot \frac{r^4}{4} \Big|_0^{\sin \psi} = \frac{\pi}{2} \int_0^1 t^4 dt = \frac{\pi}{10}$

№145 $\iiint_G f(\sqrt{x^2 + y^2 + z^2}) dx dy dz$, $G = \{\sqrt{x^2 + y^2} \leq z \leq \sqrt{2 + x^2 + y^2}\}$

$$x = r \cos \psi \cos \varphi$$

$$y = r \sin \psi \cos \varphi$$

$$z = r \sin \psi$$

$$r \cos \varphi = r \sin \psi = \sqrt{2 - r^2 \cos^2 \psi}$$

$$\cos \psi = \sin \varphi$$

$$\tan \psi \geq 1, \psi \geq \frac{\pi}{4}$$

$$r^2 \cos^2 \psi = 2 \rightarrow r \in [0, \sqrt{2}]$$

$$\iiint f(r) r^2 \cos \psi dr d\varphi d\psi =$$

$$= \int_0^{2\pi} d\varphi \int_{\pi/4}^{\pi/2} \cos \psi d\psi \int_0^{\sqrt{2}} f(r) r^2 dr = 2\pi \left(1 - \frac{1}{\sqrt{2}}\right) \int_0^{\sqrt{2}} r^2 f(r) dr =$$

$$= 2\pi \frac{2-\sqrt{2}}{2} \int_0^{\sqrt{2}} r^2 f(r) dr$$

$$\sqrt{146} \quad f(x, y, z) = x^2 + y^2, \quad G = \{(x^2 + y^2)/2 \leq z \leq 2\}$$

$$x = r \cos \varphi$$

$$y = r \sin \varphi$$

$$z = h$$

$$G = \left\{ \frac{r}{2} \leq h \leq 2 \right\}$$

$$\iiint (x^2 + y^2) dx dy dz = \int_0^{2\pi} d\varphi \int_0^2 dh \int_0^{\sqrt{2h}} r^3 dr = 2\pi \int_0^2 dh \cdot \frac{r^4}{4} \Big|_0^{\sqrt{2h}} =$$

$$= \frac{\pi}{2} \int_0^2 4h dh = 2\pi \frac{h^3}{3} \Big|_0^2 = \frac{16\pi}{3}$$

$$\sqrt{147} \quad \frac{D(x, y, z)}{D(r, \varphi, \psi)} = \begin{vmatrix} a \cos \varphi \cos \psi & -a \sin \varphi \cos \psi & 0 \\ b \sin \varphi \cos \psi & b \cos \varphi \cos \psi & 0 \\ -a \cos \varphi \sin \psi & -b \sin \varphi \sin \psi & c \cos \psi \end{vmatrix}$$

$$= c \sin \psi r^2 a b \cos \psi \sin \varphi + c r^2 \cos \psi a b$$

$$\cdot (\cos^2 \varphi \cos^2 \psi + \cos^2 \psi \sin^2 \varphi) = a b c r^2 \cos \psi (\sin^2 \varphi + \cos^2 \varphi) =$$

$$= a b c r^2 \cos \psi$$

$$\sqrt{148} \quad G = \left\{ \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1 \right\}$$

$$\iiint (x^2 + y^2) dx dy dz$$

$$0 \leq r \leq 1$$

$$0 \leq \varphi \leq 2\pi$$

$$-\frac{\pi}{2} \leq \psi \leq \frac{\pi}{2}$$

$$x = a r \cos \varphi \cos \psi$$

$$y = b r \cos \varphi \sin \psi$$

$$z = c r \sin \psi$$

$$\frac{D(x, y, z)}{D(r, \varphi, \psi)} = a b c r^2 \cos \psi$$

$$\int_0^{2\pi} d\varphi \int_{-\pi/2}^{\pi/2} d\psi \int_0^1 a b c r^4 \cos^3 \psi (b^2 + \cos^2 \varphi (a^2 - b^2)) dr = \int_{-\pi/2}^{\pi/2} d\psi \int_0^{2\pi} d\varphi \int_0^1 r^4 dr \cos^3 \psi (b^2 + \cos^2 \varphi (a^2 - b^2))$$

$$= \int_{-1}^1 (1-t^2) dt = 2 - \left(\frac{1}{3} + \frac{1}{3}\right) = \frac{4}{3}$$

$$\sqrt{6} \quad 2) (x^2 + y^2)^2 = 2a^2(x^2 - y^2), \quad x^2 + y^2 = a^2 / \sqrt{x^2 + y^2} \geq a \geq 0$$

$$r^4 = 2a^2 r^2 (\cos^2 \varphi - \sin^2 \varphi) \rightarrow r^2 = 2a^2 \cos 2\varphi$$

$$r^2 = a^2$$

$$\rightarrow \cos 2\varphi = \frac{1}{2} \rightarrow \varphi = \pm \frac{\pi}{6} + \pi n$$

$$a < r < a \sqrt{2 \cos 2\varphi}$$

$$S = 4 \int_0^{\pi/6} d\varphi \int_a^{2\cos 2\varphi} r dr = 2a^2 \int_0^{\pi/6} (2\cos 2\varphi - 1) d\varphi = \frac{3\sqrt{3} - \pi}{3} a^2$$

$$\sqrt[4]{\frac{x}{a}} + \sqrt[4]{\frac{y}{b}} = 1, x=0, y=0, a>0, b>0$$

$$x = a \cos^2 \varphi \rightarrow \sqrt[4]{r} (\cos^2 \varphi + \sin^2 \varphi) = 1 \rightarrow G = \begin{cases} 0 \leq r \leq 1 \\ 0 \leq \varphi \leq \frac{\pi}{2} \end{cases}$$

$$y = b \sin^2 \varphi$$

$$\frac{D(x, y)}{D(r, \varphi)} = \begin{vmatrix} a \cos^2 \varphi & -8ar \cos^2 \varphi \sin \varphi \\ b \sin^2 \varphi & 8br \sin^2 \varphi \cos \varphi \end{vmatrix} = 8abr \cos^2 \varphi \sin^2 \varphi$$

$$S = \iint_G dx dy = \int_0^{\pi/2} d\varphi \int_0^1 8abr \cos^2 \varphi \sin^2 \varphi dr = 8ab \int_0^1 dr$$

$$\int_0^{\pi/2} \sin^2 \varphi \cos^2 \varphi d\varphi = 4ab \cdot I$$

$$I = \frac{1}{2^7} \int_0^{\pi/2} \sin^2 \varphi d\varphi \xrightarrow{2\varphi=t} \frac{1}{2^7} \int_0^{\pi} \sin^2 t dt = \left\{ \begin{array}{l} u = \cos t \\ du = -\sin t dt \end{array} \right\} =$$

$$= \frac{1}{2^8} \int_{-1}^1 (1-u^2)^3 du = \frac{1}{2^8} \int_0^1 (1-3u^2+3u^4-u^6) du =$$

$$= \frac{1}{2^8} \left(\frac{3}{5} - \frac{1}{7} \right) \rightarrow \underline{S = \frac{ab}{70}}$$

$$\sqrt{10} \quad (a_1 x + b_1 y + c_1)^2 + (a_2 x + b_2 y + c_2)^2 = 1,$$

$$\begin{cases} u = r \cos \varphi \\ v = r \sin \varphi \end{cases}$$

$$\Delta = a_1 b_2 - b_1 a_2 \neq 0$$

$$S = \iint_G dx dy = \frac{1}{|\Delta|} \iint_G du dv = \frac{1}{|\Delta|} \iint_{G^*} r dr d\varphi = \frac{1}{|\Delta|} \int_0^{2\pi} d\varphi \int_0^1 r dr = \frac{\pi}{|\Delta|}$$

$$\sqrt{13.5} \quad z = x^2 + y^2, z = x + y$$

$$x = r \cos \varphi$$

$$y = r \sin \varphi$$

$$z = h$$

$$V = \iint_G r (a - r^2) dr d\varphi = \int_{-\frac{3\pi}{4}}^{\frac{3\pi}{4}} d\varphi \int_0^a \underbrace{r^2(a-r)}_{I_0} dr$$

$$I_0 = \frac{1}{12} a^4$$

$$V = \frac{1}{12} \int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} (\cos \varphi + \sin \varphi)^4 d\varphi = \frac{\pi}{8}$$

$$\sqrt{16.5} \quad (x^2 + y^2 + z^2)^2 = a^3 x, a > 0$$

$$x = r \cos \varphi \cos \psi$$

$$y = r \sin \varphi \cos \psi$$

$$z = r \sin \psi$$

$$r = \sqrt[3]{a^3 \cos \varphi \cos \psi}$$

$$V = \iiint_G r^2 \cos \psi d\varphi d\psi dr = \int_{-\pi/2}^{\pi/2} d\varphi \int_{-\pi/2}^{\pi/2} d\psi \int_0^{\sqrt[3]{a^3 \cos \varphi \cos \psi}} r^2 \cos \psi dr = \frac{2}{3} a^3 I_0 = \frac{\pi a^3}{3}$$

121

$$u = a_1 x + b_1 y + c_1 z$$

$$v = a_2 x + b_2 y + c_2 z$$

$$w = a_3 x + b_3 y + c_3 z$$

$$\frac{D(u, v, w)}{D(x, y, z)} = \Delta$$

$$G = \{ -d_1 < u < d_1, -d_2 < v < d_2, -d_3 < w < d_3 \}$$

$$V = \iiint_G \frac{1}{|\Delta|} du dv dw = \frac{1}{|\Delta|} \iiint_G du dv dw = \frac{8 d_1 d_2 d_3}{|\Delta|}$$

163 $\sqrt{x^2 + y^2} \leq z \leq h, \rho = \rho_0 z^2$

$$x = r \cos \varphi, \quad M_x = M_y = 0$$

$$y = r \sin \varphi$$

$$z = \lambda$$

$$M_x = \iiint_G \rho x dx dy dz = \rho_0 \iiint_G x^2 r^2 \cos \varphi r dr d\varphi d\lambda =$$

$$= \rho_0 \int_0^{2\pi} \cos \varphi d\varphi \int_0^h r^3 dr \int_0^h \lambda^2 d\lambda = 0 = M_y$$

$$M_z = \rho_0 \int_0^{2\pi} d\varphi \int_0^h r dr \int_0^h \lambda^3 d\lambda = \frac{2\pi}{4} \rho_0 \int_0^h (h^4 - r^4) r dr =$$

$$= \frac{1}{6} \pi \rho_0 h^6$$

$$M = \frac{2}{3} \pi \rho_0 \int_0^h (h^3 - r^3) r dr = \frac{2}{3} \pi \rho_0 \left(\frac{1}{2} h^3 r^2 \right)_0^h - \frac{1}{5} r^5 \Big|_0^h = \frac{1}{5} \pi \rho_0 h^5$$

$$z_c = \frac{M_z}{M} = \frac{\frac{1}{6} \pi \rho_0 h^6}{\frac{1}{5} \pi \rho_0 h^5} = \frac{5}{6} h, \quad x_c = y_c = 0$$